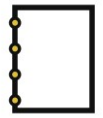
Data as tables: records, variables and keys

What exactly is one row of a table, and how does one table connect to another?

AI and Data Science Professional | Session 2 of 24 | Module A: Think with data
KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Say what one record stands for in a table (its granularity), and choose the right variable type for each variable

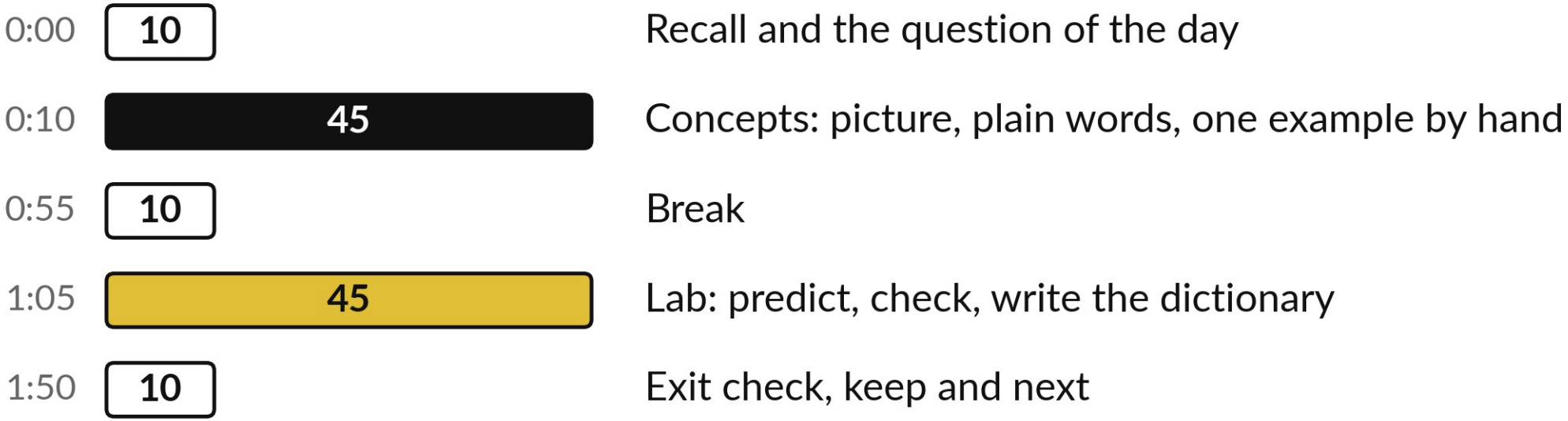


Find a primary key, a composite key and a foreign key, and read an entity relationship diagram



Test a table in Google Sheets for missing values and messy data, and write a data dictionary

Your two hours: concepts first, then a Google Sheets lab



Ideas with pictures first. Then you open the sheet. Then the checks and what you keep.

Three questions from last time

1. In CRISP-DM, which phase comes straight after business understanding, and what do you do in it?

2. In Excel words, what is a dataset, and how does a model differ from a formula?

3. An assistant gives you the share of orders paid by Mobile money. What do you do before you report it?

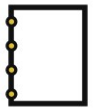


Answer aloud first. Then we check together.

Answers to the three questions from Session 1

Question	Short answer
In CRISP-DM, which phase comes straight after business understanding, and what do you do in it?	Data understanding: find the tables that could answer the question and check what they mean.
In Excel words, what is a dataset, and how does a model differ from a formula?	A dataset is a workbook in which each tab is one table. A model is a rule found from data, where a formula is typed by a person.
An assistant gives you the share of orders paid by Mobile money. What do you do before you report it?	Recompute it from your own data: 48.8 percent, 976 of 2,000. Write the source beside it.

Words we use today

	Word	What it means
	**Record (observation, instance, row)**	One row of a table: one thing the table describes. In the orders tab each record is one till order.
	**Granularity (grain)**	The level of detail of a table: one record per order, per order line or per branch per week.
	**Variable type and data type**	Variable type: how a variable is measured (numeric, categorical, datetime, text). Data type: how the cell stores it (integer, float, string). They are different ideas.
	**Primary key and foreign key**	A primary key never repeats and names one record. A foreign key holds the primary key of another table and links the two.
	**Missing value (NaN, NULL)**	A cell with no value. Ask whether it means unknown, not applicable or something real. Informative missingness: the blank itself is a fact, such as a walk-in shopper.

A table holds records in rows and variables in columns

Table: the whole grid is the orders tab

Variable (feature, field): one column

order_id	order_date	customer_id	order_value_ugx
O000003	2024-01-07		66200
O000046	2024-01-02		3100
O000054	2024-01-02	C0229	52200
O000061	2024-01-08	C0362	79830

Record (observation, instance): one row

Cell: one value, here C0362

A record is one row. A variable is one column.

Record (observation, instance): one row. Here, one till order.

Variable (feature, field): one column.

Cell: one value of one variable in one record.

The spreadsheet word and the data science word for each part

	In a spreadsheet	In data science	Plain meaning
	**Row**	Record (observation, instance)	One thing the table describes, such as one till order.
	**Column**	Variable (feature, field, attribute)	The same fact about every record, such as branch_id.
	**Cell**	Value	One variable for one record.
	**Tab**	Table (a DataFrame in pandas)	All the records of one kind.
	**Workbook**	Dataset or database	Tables that belong together.

In machine learning a variable is either a feature (an input) or the target (the answer you predict).

Granularity is the level of detail: one record per order, per order line or per branch per week

orders: each record is one till order

order_id	order_date	order_value_ugx
O000003	2024-01-07	66200

1 record

order_lines: each record is one product on one order

line_id	order_id	product_id	line_revenue_ugx
L000009	O000003	P117	58400
L000010	O000003	P043	7800

2 records

branch_weeks: each record is one branch in one week

branch_id	week_start	orders	sales_ugx
B01	2024-01-01	55	1924215

1 record

Granularity is the level of detail of a table. It says what one record stands for.

The unit of observation is what one record stands for.

Data warehouse teams call the level of detail the grain.

Finish this sentence first: each record is one ...

Each of the six tabs has its own record and its own count

	Tab	Each record is	Records	Variables
	**orders**	one till order	2,000	7
	**order_lines**	one product on one order	6,590	7
	**products**	one product	150	5
	**branches**	one branch	8	5
	**customers**	one loyalty customer	2,500	5
	**branch_weeks**	one branch in one week	832	7

The workbook holds a 2,000-order sample of the full Lakeside Mart file. Records are counted without the header row.

Counting records in the wrong tab: 6,590 lines are not 6,590 orders

Wrong

order_lines has 6,590 records.

Reporting that as the number of orders counts every product on every order.

Right

Count the orders tab: 2,000 orders.

Or count each order_id once in order_lines: 2,000.

That is 3.3 lines per order on average.

State the granularity first. Then count.

Every variable has a variable type (how it is measured) and a data type (how it is stored)

Variable type: how a value is measured

Numeric	Categorical	Datetime	Text
Discrete: counted e.g. quantity	**Nominal: no order** e.g. payment_method	**A date or a time** e.g. order_date	**Free words** e.g. message in feedback.csv
Continuous: measured e.g. order_value_ugx, in UGX	**Ordinal: has an order** e.g. loyalty_tier: Basic, Silver, Gold	**Sorts in time order** e.g. week_start	**Needs special tools** We meet them in Session 21

Data type: how the cell stores the value

integer or float	string, or an integer code	datetime (here: a string)	string

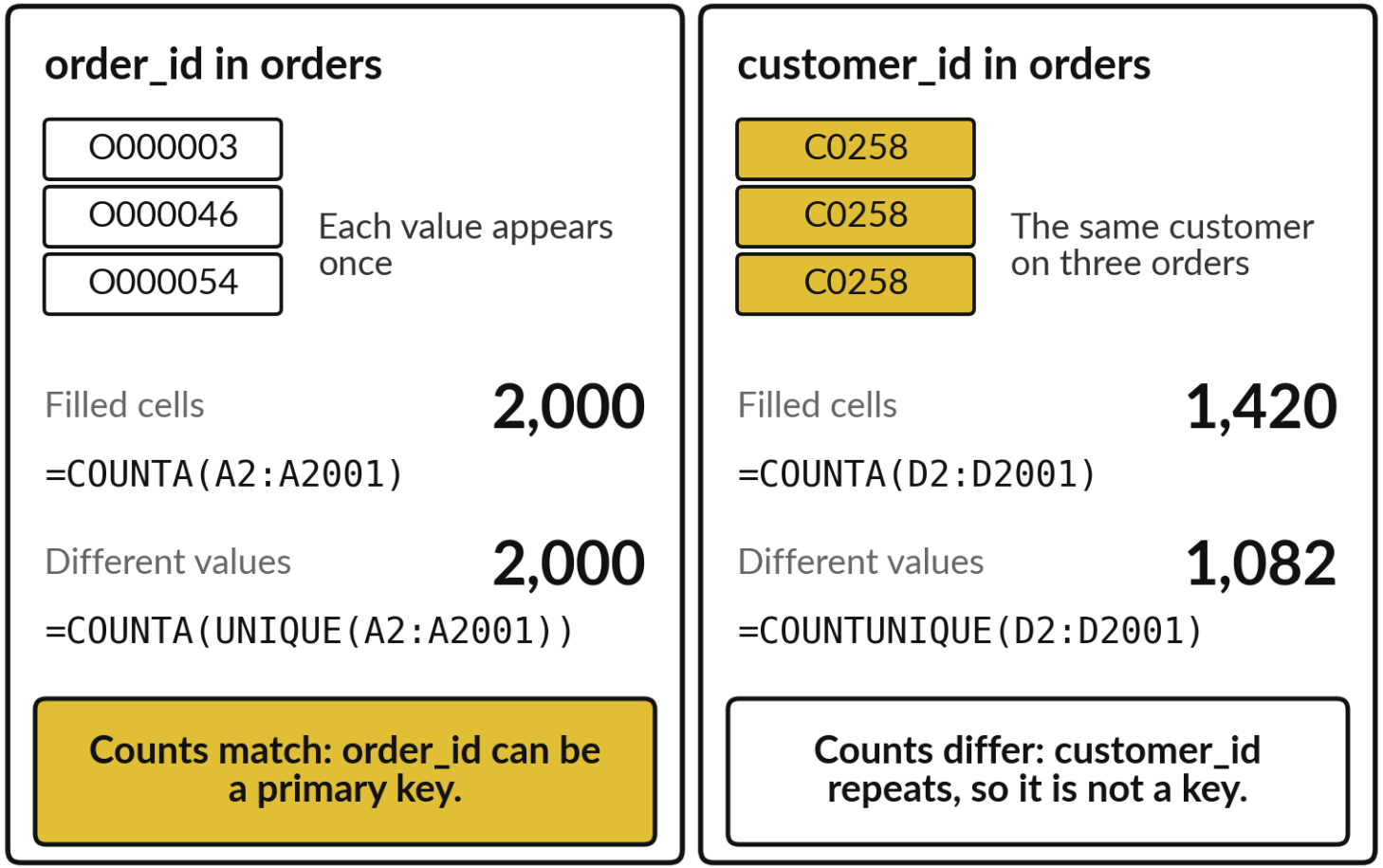
Discrete: counted. Continuous: measured. An identifier (order_id) is a label: never add it up.

The seven variables of the orders tab and their types

	Variable	Example (order O000054)	Data type in this workbook	Variable type
	**order_id**	O000054	string	identifier (a label)
	**order_date**	2024-01-02	string: not a real date	datetime
	**branch_id**	B01	string	categorical (nominal)
	**customer_id**	C0229	string, 580 blank	identifier, with missing values
	**payment_method**	Cash	string	categorical (nominal)
	**promo_week**	0	integer	categorical (a 0 or 1 flag)
	**order_value_ugx**	52200	integer	numeric (continuous, UGX)

An identifier can look like a number or a code. You never add it up.

A primary key is a variable that is different in every record



Primary key: a variable whose value is different in every record, so it names exactly one record.

Test it: count the filled cells, then count the different values.

If the two counts match, it can be a primary key.

order_id passes. customer_id repeats, so it fails.

Two variables together can form a composite key

branch_id	week_start	orders	sales_ugx
B01	2024-01-01	55	1924215
B01	2024-01-08	51	1529300
B01	2024-01-15	66	3296170
B02	2024-01-01	70	3397835
B02	2024-01-08	42	1645550
B02	2024-01-15	42	1935690

B01 repeats: it appears 3 times here.

week_start repeats: 2024-01-01 appears for both branches.

The pair (B01, 2024-01-01) appears once, so the pair is the key.

8 branches × 104 weeks = 832 records

Composite key: two or more variables that are unique together, although each repeats alone.

branch_id repeats: every branch appears once per week.

week_start repeats: every week appears once per branch.

Each pair appears once, so the pair is the key: 832 records.

A foreign key holds the primary key of another table and links the two

orders: the many side

FK

order_id	branch_id
O000003	B01
O005886	B02
O014061	B04

branches: the one side

PK

branch_id	branch_name
B01	Kampala Central
B02	Kampala Ntinda
B03	Entebbe
B04	Jinja

```
=VLOOKUP("B01", branches!A2:B9, 2, FALSE)
```

The lookup returns Kampala Central.

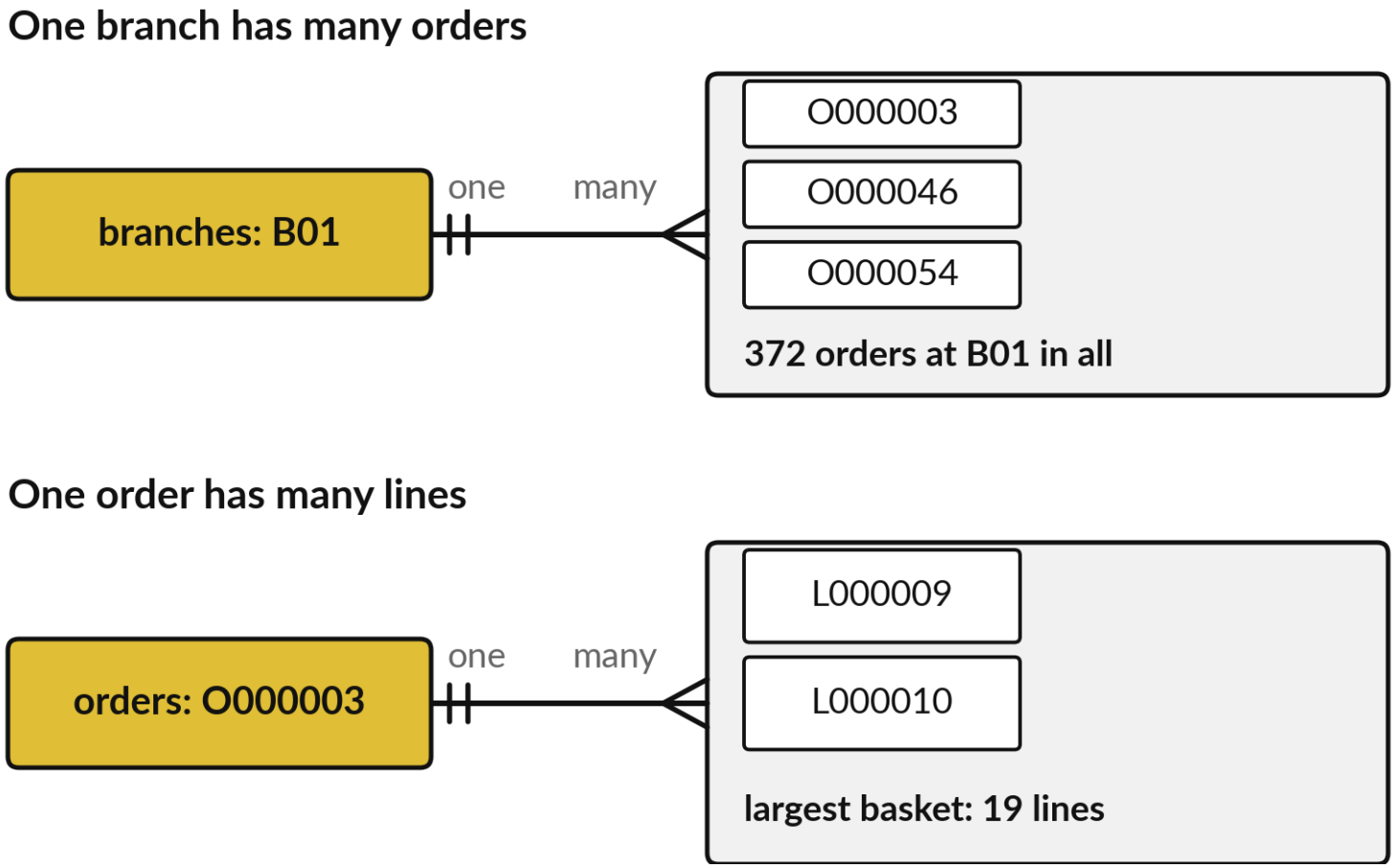
Foreign key: a variable whose values are primary keys of another table.

branch_id in orders is a foreign key. branch_id in branches is the primary key.

Look up B01 in branches to find the branch name.

In SQL and pandas this lookup is called a join or a merge.

Tables relate one-to-many: one branch has many orders and one order has many lines



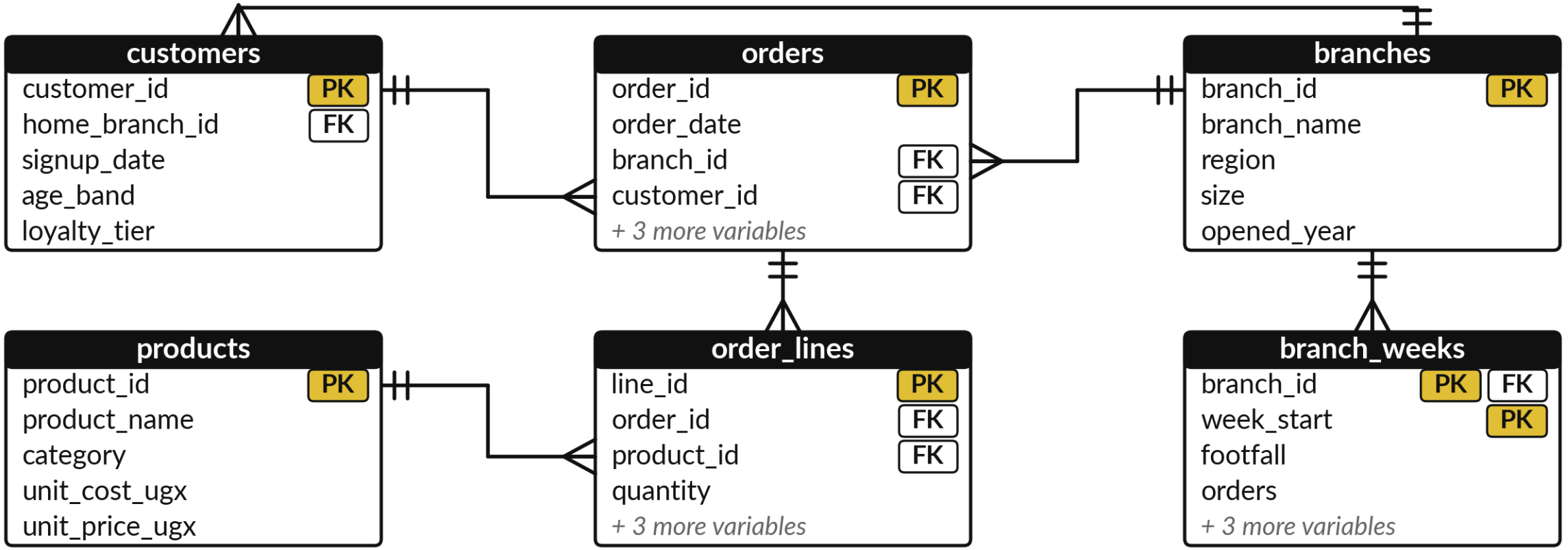
A relationship says how records of two tables match. The common kind is one-to-many.

One branch has many orders: B01 has 372.

One order has many lines: O000003 has 2.

Many-to-many: orders and products, joined by the junction table order_lines.

An entity relationship diagram (ERD) draws the six tabs and how their keys link



PK marks a primary key and FK a foreign key. A line ends in three prongs on its many side.

Follow order O000003 through four tabs by their keys

orders: one record

order_id	O000003
branch_id	B01
customer_id	(blank)
order_value_ugx	66200

1 order_id

order_lines: two records for this order

line_id	product_id	quantity	line_revenue_ugx
L000009	P117	4	58400
L000010	P043	1	7800

2 product_id

3 branch_id

branches: one record

branch_id	B01
branch_name	Kampala Central
region	Central

products: one record for each product

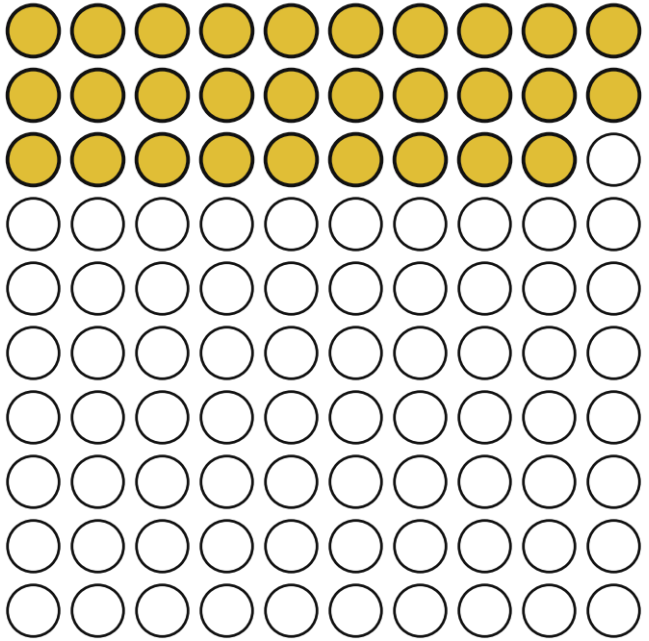
product_id	product_name	unit_price_ugx
P117	Household item 117	14600
P043	Staples item 043	7800

customer_id is blank, so this order has no row in customers: it was a walk-in.

A blank customer_id is a missing value, and it appears on 29 percent of orders

29%

580 of the 2,000 orders have no customer_id. The data dictionary says the blank means a walk-in: a shopper with no loyalty number.



orders in every 100 have a blank customer_id

A blank can mean unknown, not applicable or a walk-in, and each needs different handling

	What the blank means	Example	What to do
	**Unknown: the value exists but was not recorded**	loans.csv: 379 of 8,000 loans have no monthly_income_ugx	Impute (fill in) with the median, and add a missing-indicator flag
	**Not applicable: no value can exist**	Illustrative: a refund reason on an order that was never refunded	Leave it blank, or write none
	**Informative missingness: the blank itself is a fact**	orders: 580 blank customer_id values mean a walk-in	Keep it, and add a flag such as is_walk_in

Names for a blank: NaN in pandas, NULL in SQL, an empty cell in Sheets.

A raw export is messy: three payment methods appear under 11 spellings

	order_date	branch	payment	qty	unit_price
2	2024-10-01	Gulu	Mobile money	2.0	7000
3	11/03/2024	Fort Portal	Mobile money	1.0	3100
4	2024-08-25	Kampala Cntral	Cash	1.0	30100
5	2025-03-26	Entebbe	MM	2.0	2000
6	2025-07-11	Gulu	Mobile money		2100
7	2024-08-27	Mbale	Mobile money	1.0	UGX 1,700

Row 3 Date written day/month/year. Most dates in the file are year-month-day.

Row 4 Branch name misspelt: Kampala Cntral.

Row 5 MM is short for Mobile money.

Row 6 Quantity is missing.

Row 7 Price stored as text, with a UGX prefix and a comma.

A raw order-line export (orders_raw.csv). Gold cells are the problems. Session 8 cleans this file.

Six messy-data checks you can run in Sheets before you trust a total

Problem	What we found	First check in Sheets
Blank cells	customer_id: 580 blank in orders	=COUNTBLANK(D2:D2001)
Dates stored as text	order_date in orders is text	=ISNUMBER(B2) gives FALSE
Inconsistent labels	11 spellings of 3 payment methods (raw export)	=COUNTUNIQUE(range) in Sheets
Extreme values	Largest order 1,935,520 UGX, median 24,800	=MAX(G2:G2001) and =MEDIAN(G2:G2001)
Duplicate records	80 exact duplicate lines (raw export)	=ROWS(range)-ROWS(UNIQUE(range)) in Sheets
Numbers stored as text	UGX 1,700 in unit_price (raw export)	=ISNUMBER(cell)

A data dictionary says what each variable means, its type and its key

	A	B	C	D
	variable	**variable type**	**key**	**meaning**
1	order_id	identifier	primary key	One till order
2	order_date	datetime		Date of the order
3	branch_id	categorical	foreign key	Branch that sold it
4	customer_id	identifier	foreign key	Loyalty customer; blank for a walk-in
5	payment_method	categorical		Cash, Mobile money or Card
6	promo_week	categorical		1 if the branch ran a promotion that week
7	order_value_ugx	numeric		Sum of the order lines, in UGX

The orders tab, written as a data dictionary. You write one for every tab in the lab.

Where you will use this at work: check keys and granularity before you report

	Your job	The tables you meet	What today lets you do
	**Credit analyst at a bank in Kampala, using SQL**	loans: one record per loan (loans.csv).	Check loan_id is unique before you join it to anything, so no loan is counted twice.
	**Health records officer at a district office, using pandas**	Monthly clinic reports: 18 of 720 district-months blank in our file.	Tell a blank (not reported) from a zero before you total a district.
	**Sales analyst handing a workbook to Power BI**	Orders, order lines and branches.	Mark each key first, so the relationships in Power BI do not multiply totals.
	**Officer at a SACCO or coffee cooperative, using Excel**	Member register, deliveries and payments.	Draw the ERD and write the data dictionary, so a new colleague can read the workbook.

Read the recipe before the lab: five formulas test a table

```
=COUNTA(A2:A2001)
=COUNTA(UNIQUE(A2:A2001))
=COUNTBLANK(D2:D2001)
=COUNTIF(E2:E2001,"Mobile money")
=ISNUMBER(B2)
```

In plain words

Line 1 counts the filled cells of order_id: 2,000 records.

Line 2 counts the different values. Equal to line 1 means a primary key.

Line 3 counts the empty cells of customer_id: 580.

Line 4 counts the orders paid by Mobile money: 976.

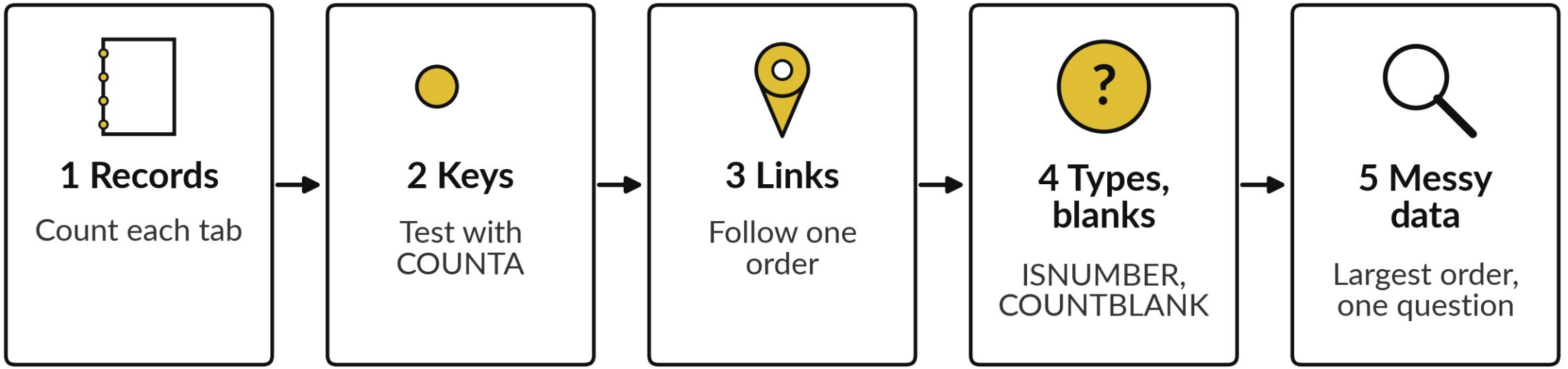
Line 5 is TRUE for a real date or number. Here it is FALSE: the date is text.

KLAI

Part 2: the Sheets lab

Close the slides. Open the sheet.

The lab has five steps, and you keep a dictionary tab at the end



Ten tasks in 45 minutes. For each task, predict the answer, check it with a formula, then write it on your dictionary tab.

Check your numbers before you move on

What you compute	You should get
Records in the six tabs, in the order of the table on slide 10	2,000, 6,590, 150, 8, 2,500, 832
Different order_id values in orders	2,000: a primary key
Lines per order on average, and in O000003	3.3, and 2
Different branch_id, week_start and pairs in branch_weeks	8, 104 and 832
Blank customer_id in orders, and its share	580, which is 29 percent
Different customer_id values among the 1,420 filled	1,082
ISNUMBER of an order_date	FALSE
Largest order_value_ugx, and the median	1,935,520 and 24,800

If your number differs, find the task that went wrong. Do not change a number to match.

Five likely mistakes in the lab and how to fix them

	What you see	What it means	What to do
✗	**A count of 2,001 where you expected 2,000**	The range includes the header cell in row 1.	Start the range in row 2: A2:A2001.
✗	**#REF! or #NAME? when you point to another tab**	The tab name is missing or misspelt.	Write the tab, then !, then the range: orders!A2:A2001.
✗	**#N/A from VLOOKUP**	The key was not found, or FALSE is missing.	Add FALSE as the last argument. Check for stray spaces.
✗	**Array result was not expanded**	Cells below the UNIQUE formula hold data.	Clear them, or move the formula to an empty column.
✓	**ISNUMBER on a date gives FALSE**	That is correct: the date is stored as text.	Write it on your dictionary tab as a finding.

Exit check: three questions, then compare with the short answers

	Question	Short answer
	**Order O000003 has 2 products. How many records does it make in orders, and how many in order_lines?**	One record in orders and 2 in order_lines. The two tabs have different granularity.
	**Which variable is the primary key of products, and which tab holds it as a foreign key?**	product_id. The order_lines tab holds product_id as a foreign key.
	**customer_id is blank on 580 orders. What do you do first?**	Read the data dictionary: the blank means a walk-in. Keep the records and add a flag, instead of deleting or filling them.

Save your workbook with its dictionary tab

What you keep from today

Keep your dictionary tab: for each of the six Lakeside tabs, one sentence saying what a record is, its key, and the problems you found, each with a number.

Our question today

What exactly is one row of a table, and how does one table connect to another?

Next session

Session 3: Summarising and showing data fairly. We describe thousands of records with a few numbers and one chart.

Take away

Your lab file

A dictionary tab

The key test

Two counts match

Next session

Summarising data